Low Level Design (LLD)

Supply Chain Management

Revision Number - 1.2

Last Date of Revision - 20/03/2025

Shubham tembhurne

Document Control

Date	Version	Description	Author
10/03/2025	1.0	Introduction, Problem Statement	Shubham Tembhurne
10/03/2025	1.1	Dataset Information, Architecture Description	Shubham Tembhurne
20/03/2025	1.2	Final Revision	Shubham Tembhurne

Contents

Document Version Control 2
1. Introduction 4
1.1 What is Low-Level Design Document? 4
1.2 Scope 4
1.3 Project Introduction 4
2. Problem Statement 5
3. Dataset Information 5
4. Architecture 6
4.1 Architecture Description 6

1. Introduction

1.1 What is Low-Level Design Document?

The goal of the Low-level design document (LLDD) is to give the internal logic design of the actual program code for the Supply Chain Analysis dashboard. LLDD describes the class diagrams with the methods and relations between classes and programs specs. It describes the modules so that the programmer can directly code the program from the document.

2. What is Scope?

Low-level design (LLD) is a component-level design process that follows a step-by-step refinement process. The process can be used for designing data structures, required software architecture, source code, and ultimately, performance algorithms. Overall, the data organization may be defined during requirement analysis and then refined during data design work.

3. Project Introduction

AtliQ Mart is a growing FMCG (fast moving consumer Good) manufacturer headquartered in Gujarat, India. It is currently operational in three cities Vadodara, Surat and Ahmedabad. They want to expand to other metro/tier1 cities in the next 2 years.

2. Problem Statement

AtliQ Mart is currently facing a problem where a few key customers did not extend the annual contract due to service issues. It is speculated that some of the essential products were either not delivered on time or not delivered in full over a continued period, which could have resulted in bad customer service Management wants to fix this issue before expanding to other cities and requested their supply chain analytics team to track the 'On time' and 'In Full' delivery service level for all the customers on a daily basis so that they can respond swiftly to these issues. The Supply Chain team decided to use a standard approach to measure the service level in which they will measure 'on-time delivery (OT) %', 'In-full delivery (IF) %' and On Time in full (OTIF) % of the customer orders on a daily basis against the target service level set for each customer

3. Dataset Information

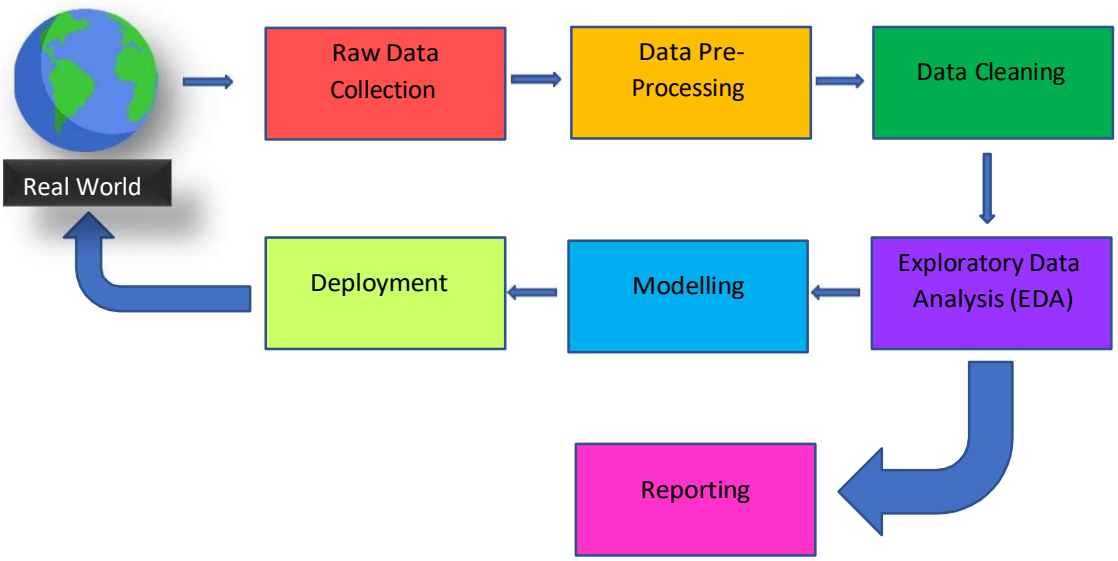
Column Description for fact_order_lines: This table contains all information about orders and each item inside the orders.

1. order_id: Unique ID for each order the customer placed
2. order_placement_date: It is the date when the customer placed the order
3. customer_id: Unique ID that is given to each of the customers
4. product_id: Unique ID that is given to each of the products
5. order_qty: It is the number of products requested by the customer to be delivered
6. agreed_delivery_date: It is the date agreed between the customer and AtliQ Mart to deliver the products
7. actual_delivery_date: It is the actual date AtliQ Mart delivered the product to the customer
8. delivered_qty: It is the number of products that are actually delivered to the customer

Column Description for fact_orders_aggregate: This table contains information about OnTime, InFull and OnTime Infull information aggregated at the order level per customer

1. order_id: Unique ID for each order the customer placed
2. customer_id: Unique ID that is given to each of the customers
3. order_placement_date: It is the date when the customer placed the order
4. on_time: '1' denotes the order is delivered on time. '0' denotes the order is not delivered on time.
5. in_full: '1' denotes the order is delivered in full quantity. '0' denotes the order is not delivered in full quantity.
6: otif: '1' denotes the order is delivered both on time and in full quantity. '0' denotes the order is either not delivered on time or not in full quantity

4. Architecture



1. Architecture Description

1. **Raw Data Collection**
   The Dataset was taken from CloudyML Provided Project Description Document.
   https://drive.google.com/drive/folders/1AkVENpnXau9MdW87bVmLtuBFap8R2i6m

2. **Data Pre-Processing**
   Before building any model, it is crucial to perform data pre-processing to feed the correct data to the model to learn and predict. Model performance depends on the quality of data fed to the model to train.
   This Process includes-
   a) Handling Null/Missing Values
   b) Handling Skewed Data
   c) Outliers Detection and Removal

3. **Data Cleaning**
   Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset.
   a) Remove duplicate or irrelevant observations
   b) Filter unwanted outliers
   c) Renaming required attributes

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis refers to the critical process of performing initial investigations on data to discover patterns, spot anomalies, test hypotheses, and check assumptions with the help of summary statistics and graphical representations.

5. Reporting

Reporting is a most important and underrated skill of a data analytics field. Because being a Data Analyst you should be good in the easy and self-explanatory report because your model will be used by many stakeholders who are not from a technical background.

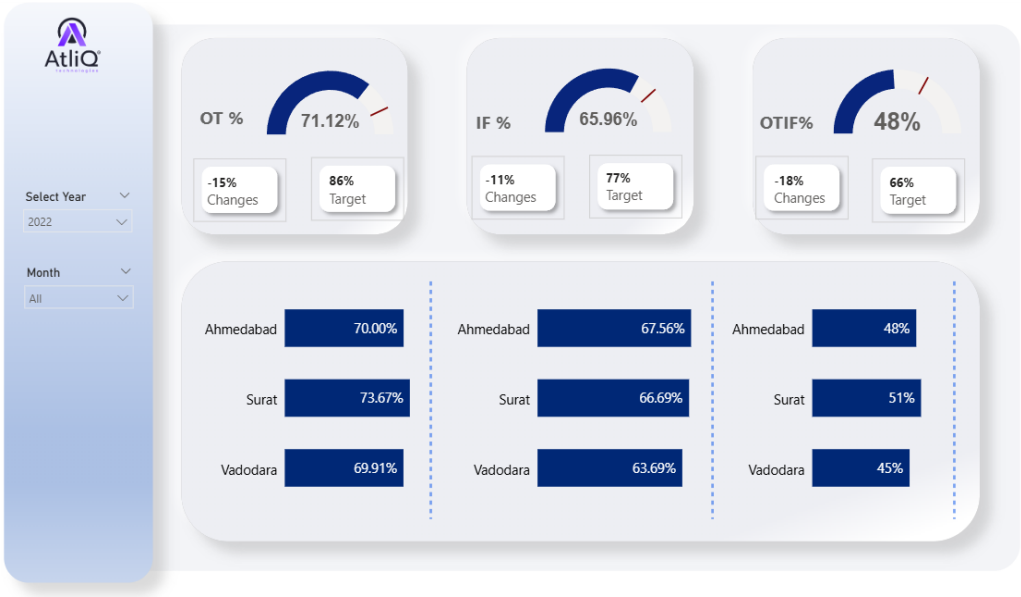
a) High-Level Design Document (HLD)
b) Low-Level Design Document (LLD)
c) Architecture
d) Wireframe
e) Detailed Project Report
f) PowerPoint Presentation

6. Modelling

Data Modelling is the process of analyzing the data objects and their relationship to the other objects. It is used to analyze the data requirements that are required for the business processes. The data models are created for the data to be stored in a database. The Data Model's main focus is on what data is needed and how we have to organize data rather than what operations we have to perform.

7. Deployment

We created a Power BI Dashboard



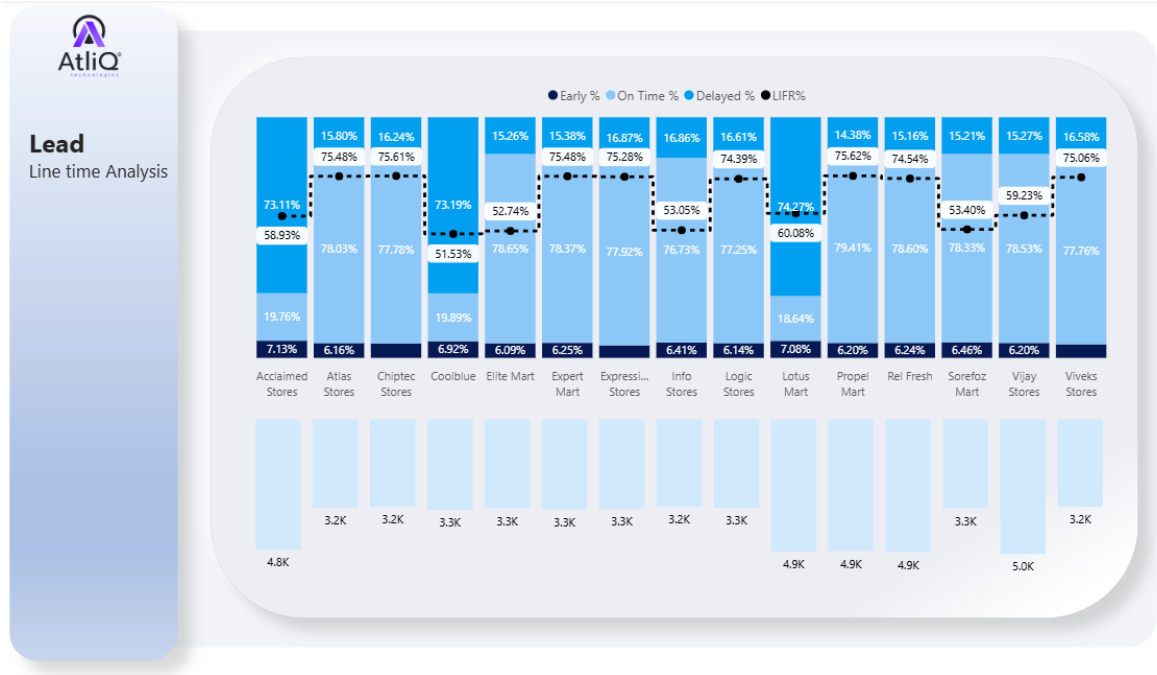
Metric	Acclaimed Stores	Atlas Stores	Chiptec Stores	Coolblue	Elite Mart	Expert Mart	Expression Stores	Info Stores	Logic Stores	Lotus Mart	Propel Mart	Rel Fresh	Sorefoz Mart	Vijay Stores	Viveks Stores
IF%	58.93%	75.48%	75.61%	51.53%	52.74%	75.48%	75.28%	53.05%	74.39%	60.08%	75.62%	74.54%	53.40%	59.23%	75.06%
OT%	26.89%	84.20%	83.76%	26.81%	84.74%	84.62%	83.13%	83.14%	83.39%	25.73%	85.62%	84.84%	84.79%	84.73%	83.42%
OTIF%	15.24%	63.80%	62.97%	13.75%	44.88%	63.61%	62.31%	43.48%	61.96%	16.06%	65.05%	63.04%	45.60%	49.61%	63.04%

AtliQ
Trends
for
Key Metrics
VOFR%
97.0%
96.5%
96.50%
96.59%
96.54%
96.66%
96.64%
96.74%
96.48%
96.37%
96.51%
96.70%
96.51%
96.58%
96.56%
96.75%
96.45%
96.59%
96.61%
96.39%
96.53%
96.53%
96.62%
96.88%
96.64%
96.67%
96.45%
96.74%
96.62%
W 10 W 11 W 12 W 13 W 14 W 15 W 16 W 17 W 18 W 19 W 20 W 21 W 22 W 23 W 24 W 25 W 26 W 27 W 28 W 29 W 30 W 31 W 32 W 33 W 34 W 35 W 36
96.85% 96.64% 96.87% 96.75% 96.80% 96.77%
96.56% 96.62% 96.56% 96.71% 96.70% 96.75%
96.32% 96.37% 96.35% 96.19% 96.37% 96.41%
March April May June July August
2022
Ahmedabad Surat Vadodara
96.68% 96.62% 96.62% 96.65% 96.65% 96.66%
96.58% 96.53% 96.61% 96.54% 96.57% 96.65%
96.55% 96.51% 96.59% 96.44% 96.56% 96.55%
March April May June July August
Beverages Dairy Food

AtliQ
Order Line
At Glance
96.59%
VOFR%
65.96%
LIFR%
57K
Total Order Line
12969K
Total Order LIne QTY
Deliverd
38K
Total Order Line In Full
13427K
Total Order LIne QTY
100%
50%
0%
Viveks Stores
Vijay Stores
Sorefoz Mart
Rel Fresh
Propel Mart
Lotus Mart
Logic Stores
Info Stores
Express... Stores
Expert Mart
Elite Mart
Coolblue
Chiptec Stores
Atlas Stores
Acclai... Stores
Total order Line
LIFR%

Delayed Days	Beverages	Dairy	Food	Total
-1	584	2454	569	3607
0	6171	24521	6306	36998
1	1377	5557	1321	8255
2	843	3452	841	5136
3	486	2112	502	3100
Total	9461	38096	9539	57096

Delayed Days	Ahmedabad	Surat	Vadodara	Total
-1	1243	1094	1270	3607
0	12531	12050	12417	36998
1	2934	2389	2932	8255
2	1819	1458	1859	5136
3	1149	851	1100	3100
Total	19676	17842	19578	57096



category	product_name	LIFR%	VOFR%	LIFR% by Date	VOFR% by Date
Beverages	AM Tea 100	65.90%	96.63%		
	AM Tea 250	65.34%	96.52%		
	AM Tea 500	65.12%	96.48%		
Dairy	AM Butter 100	66.17%	96.62%		
	AM Butter 250	64.25%	96.38%		
	AM Butter 500	65.64%	96.59%		
	AM Curd 100	66.25%	96.58%		
	AM Curd 250	66.30%	96.61%		
	AM Curd 50	65.57%	96.58%		
	AM Ghee 100	65.96%	96.57%		
	AM Ghee 150	66.46%	96.68%		
	AM Ghee 250	65.50%	96.53%		
	AM Milk 100	66.46%	96.63%		
	AM Milk 250	65.63%	96.62%		
	AM Milk 500	66.85%	96.63%		
Food	AM Biscuits 250	66.41%	96.64%		
	AM Biscuits 500	66.74%	96.62%		
	AM Biscuits 750	66.73%	96.63%		